

FACIAL EXPRESSION RECOGNISATION SYSTEM USING CNN ALGORITHM AND DL

COLLEGE MINOR PROJECT — 5th SEMESTER

UNDER THE GUIDANCE:
MS. ZEBA RUQSHANH
(ASSISTANT PROFESSOR)

PRESENTED BY:
SYED MOAZAM
(160923748131)

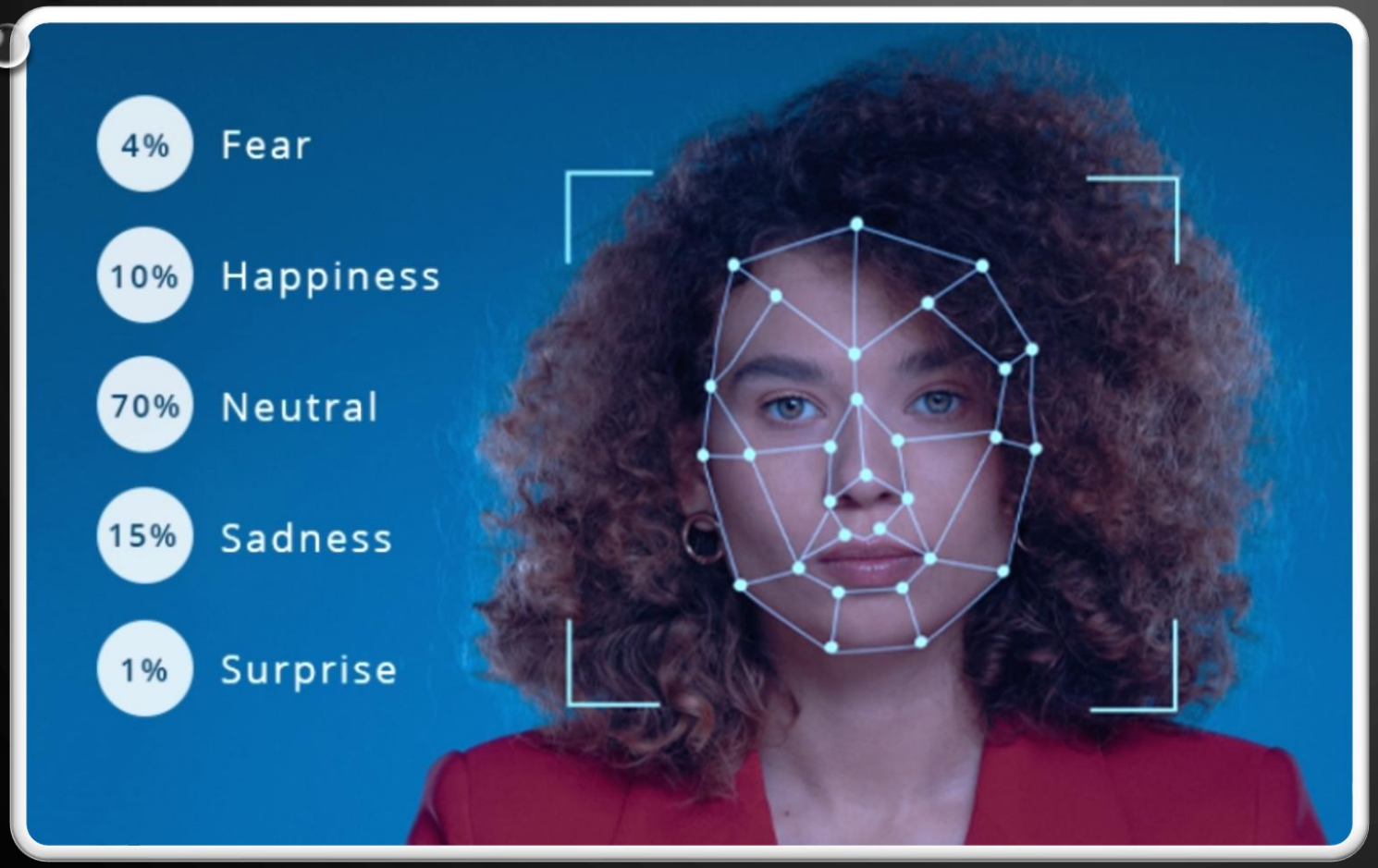
ABSTRACT

Facial Expression Recognition (FER) refers to the technology that can automatically detect and interpret facial expressions from images or video frames.

Emotions like happiness, sadness, anger, fear, surprise, and disgust are commonly studied in FER.

Deep Learning is a subset of machine learning that uses neural networks with multiple layers (deep architectures) to learn complex patterns from data.

CNNs are a class of deep neural networks specifically designed for image analysis. They automatically detect spatial hierarchies in visual data.



OBJECTIVES

- AUTOMATE EMOTION DETECTION BY APPLYING IMAGE PREPROCESSING AND CNN-BASED FEATURE EXTRACTION FOR ACCURATE FACIAL PATTERN ANALYSIS.
- TRAIN AND VALIDATE A DEEP LEARNING MODEL TO CLASSIFY A VARIETY OF HUMAN EMOTIONS EFFECTIVELY.
- IMPROVE MODEL ROBUSTNESS TO HANDLE REAL-WORLD CHALLENGES SUCH AS LIGHTING CHANGES, OCCLUSIONS, AND NATURAL EXPRESSION VARIATIONS.

EXISTING SYSTEM

EXISTING FACIAL EXPRESSION RECOGNITION SYSTEMS PRIMARILY USE TRADITIONAL MACHINE LEARNING ALGORITHMS WITH MANUALLY EXTRACTED FEATURES (LIKE LBP, HOG). THESE SYSTEMS ARE LIMITED IN ACCURACY, SCALABILITY, AND PERFORMANCE UNDER REAL-WORLD CONDITIONS.

1. TRADITIONAL MACHINE LEARNING APPROACHES
2. FEATURE EXTRACTION + CLASSIFICATION PIPELINE
3. BASIC NEURAL NETWORKS

DRAWBACKS

- **REQUIRES LARGE DATASETS:** NEEDS EXTENSIVE LABELED FACIAL EXPRESSION DATA FOR ACCURATE TRAINING.

- **HIGH COMPUTATIONAL POWER:** TRAINING CNN MODELS DEMANDS POWERFUL GPUS, WHICH CAN BE COSTLY.

- **RISK OF OVERFITTING:** CNN MODELS MAY OVERFIT WITHOUT PROPER REGULARIZATION AND DATA AUGMENTATION.

- **REAL-TIME CHALLENGES:** LIMITED HARDWARE CAN CAUSE DELAYS IN REAL-TIME EXPRESSION RECOGNITION.

- **COMPLEX IMPLEMENTATION:** DEVELOPING AND TUNING DEEP LEARNING MODELS REQUIRES EXPERTISE.

- **PRIVACY CONCERNS:** USE OF FACIAL DATA RAISES ETHICAL AND PRIVACY ISSUES.

PROPOSED SYSTEM AND ADVANTAGES



THE PROPOSED SYSTEM COMBINES **YOLO (YOU ONLY LOOK ONCE)** FOR FAST AND ACCURATE FACE DETECTION WITH A **CNN-BASED DEEP LEARNING MODEL** FOR AUTOMATIC FACIAL EXPRESSION RECOGNITION, ENABLING REAL-TIME AND ROBUST EMOTION CLASSIFICATION.

ADVANTAGES

- AUTOMATIC FEATURE LEARNING
- HIGHER ACCURACY
- ROBUST TO VARIATIONS
- SCALABLE
- POTENTIAL FOR REAL-TIME PROCESSING
- REDUCED MANUAL INTERVENTION

SYSTEM REQUIREMENTS

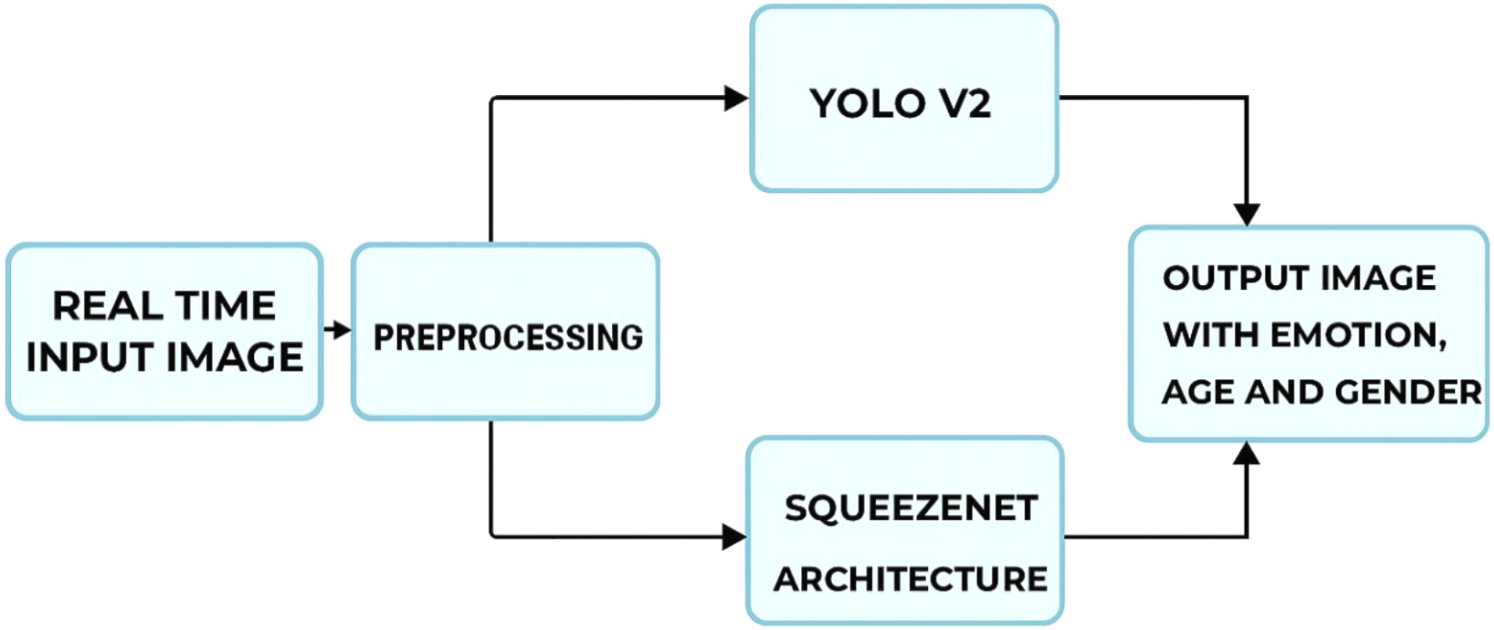
SOFTWARE REQUIREMENTS

- Programming Language: Python
- Deep Learning Libraries: TensorFlow, Keras, or PyTorch
- Image Processing: OpenCV, NumPy
- IDE: Jupyter Notebook, VS Code, or PyCharm

HARDWARE REQUIREMENTS

- CPU: Multi-core processor (Intel i5 or higher recommended)
- GPU: NVIDIA GPU with CUDA support (e.g., GTX 1050 or higher) for faster training
- RAM: Minimum 8 GB (16 GB recommended for large datasets)
- Storage: Sufficient space for datasets and model checkpoints (at least 50 GB)

SYSTEM ARCHITECTURE



CONCLUSION

Facial Expression Recognition (FER) plays a crucial role in enhancing human-computer interaction, security systems, and healthcare diagnostics.

The use of **Convolutional Neural Networks (CNNs)** and **Deep Learning** significantly improves the **accuracy** and **efficiency** of emotion detection from facial images.

It successfully classifies key expressions like happiness, sadness, anger, surprise, etc., with promising performance metrics.

The integration of CNNs allows for **automatic feature extraction**, reducing the need for manual preprocessing.

With further training on larger and more diverse datasets, the system can be made even more **robust, real-time,** and **applicable across various domains**.